

In-season cotton yield forecasting using high resolution satellite imagery

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Abstract:

Accurate in-season cotton yield forecasting at fine spatial resolution remains a challenge due to the crop's physiological complexity, the limitations of UAS-based monitoring at scale, and the inadequacy of existing satellite-based methods that rely on single-date or magnitude-only vegetation indices. This study developed Pheno-SST, a hybrid deep learning architecture combining a Transformer encoder and a Mamba state-space branch. This model trained on near-daily PlanetScope NDVI time series and its first-order derivative on years 2021-2023 and tested on 2024 growth season in South Texas, USA. The first-order derivative, encoding the rate and direction of canopy change, was critical for cross year model generalization. Models trained on NDVI magnitude alone overfitted on the unseen 2024 test year, while Pheno-SST with derivative input achieved an RMSE of 2.89 and a near-zero bias of +0.30, outperforming all competing architectures including BiLSTM, PatchTST, Random Forest, TCN, and DLinear. Notably, the 2024 season was statistically distinct from training years in peak timing, canopy persistence, and cumulative growth, yet the model generalized effectively, demonstrating robustness to environmental heterogeneity. Conformal prediction intervals confirmed well-calibrated uncertainty, with 89.6% and 92.5% coverage at the 90% and 95% confidence levels, respectively. In-season forecasting showed that reliable predictions require the crop to reach approximately 105 to 117 days after planting, consistent with the onset of reproductive-stage dynamics in cotton. These results establish a scalable, physiologically informed, and uncertainty-aware framework for operational satellite-based cotton yield forecasting.

Introduction:

Aerial remote sensing has emerged as a powerful tool for estimating yield in row crops, with substantial progress over the past decade driven largely by the uncrewed aerial systems (UAS). UAS platforms offer the capability to collect high spatial resolution imagery that, through further processing, enables detailed characterization of within-field variability in crop canopy structure, vigor, and stress. (Yang et al., 2025; El Melki et al., 2026). Studies across wheat, corn, soybean, cotton and rice have demonstrated strong correlations of UAS-derived crop canopy features including Canopy Height (CH), Canopy Cover (CC), and Vegetation Indices (VIs) with grain yield (Stanton et al., 2017; Bhandari et al., 2021; Herr et al., 2023; Zhang et al., 2023; Sarkar et al., 2025). In cotton, UAS-based multispectral and RGB imaging has proven effective for

estimating leaf area index, plant height, biomass, and ultimately lint yield (Ashapure et al., 2020; Zhao et al., 2024; Ghansah et al., 2025). Despite these advantages, UAS deployment faces several operational constraints that limit its scalability for large-area coverage and regional yield forecasting. Adverse weather conditions such as wind speeds above 10 m s⁻¹ (Campbell et al., 2017; Averyanova et al., 2021), regulatory restrictions on flight altitude and line-of-sight requirements (Matalonga et al., 2022) limits practical coverage for high acreage farms. For cotton production regions spanning thousands of hectares, achieving consistent temporal coverage across multiple farms throughout a 120–150 day growing season becomes logistically prohibitive and economically impractical (Zhao, 2025). Battery endurance, data storage, processing pipelines, and the need for trained pilots further add layers of complexity that hinder routine, systematic monitoring at landscape scales.

Satellite remote sensing offers a complementary solution by providing systematic, repeatable coverage over large areas with minimal ground infrastructure (Pettorelli et al., 2014). Additionally, recent advances in satellite technology have dramatically improved both spatial and temporal resolution. This has narrowed the historical gap between satellite and UAS capabilities. For example, Commercial constellations such as PlanetScope offer near-daily 3m imagery, with the newer SuperDove sensors incorporating red-edge channels that bridge the resolution gap between Sentinel-2 and traditional UAS platforms (Kabir et al., 2025). These satellite systems have enabled robust yield forecasting across multiple crops by capturing their short phenological stages due to its high temporal frequency. For example, in corn, almost daily 3m spatial resolution imagery reduced yield prediction error by 15% compared to Sentinel-2 alone by better resolving the heading and grain-filling periods (Li et al., 2022). Similar conclusions were also derived on rice and wheat studies (Zhao et al., 2020; San Bautista et al., 2022). These advances suggest that modern satellite constellations with a range of sensors can support operational yield forecasting without the logistical burdens of UAS deployment.

Cotton (*Gossypium hirsutum* L.) is a globally significant fiber and oilseed crop, cultivated on approximately 29.5 million hectares worldwide during 2025-26 and supporting livelihoods for over 100 million families and contributing \$600 billion to global economy (Fan et al., 2026). Global cotton markets are characterized by price volatility driven by fluctuations in supply, demand, weather shocks, and geopolitical factors (Kalli et al., 2025; Soni et al., 2025). Therefore, timely and accurate in-season yield forecasts are critical for multiple stakeholders. Yield forecasts help growers manage inputs efficiently and increase profits. Ginners and merchants use regional forecasts to plan procurement and processing to minimize supply-chain risks. Policymakers and traders rely on early forecasts to guide market decisions, insurance, and trade actions (Schnepf, 2017; Meroni et al., 2021; Betew et al., 2025).

Despite this clear demand, operational cotton yield forecasting remains less mature and less standardized than for major grain crops (Johnson, 2014; Jones et al., 2017). Approaches range from county or regional level statistical models based on historical trends and weather indices (Johnson, 2014), to process-based crop simulation models such as GOSSYM and CROPGRO Cotton (Abbas et al., 2020; Wang et al., 2020; Mishra et al., 2021), to more recent machine learning

frameworks integrating satellite imagery (Lang et al., 2023; Shahzad et al., 2025). However, few of these methods are integrated into user-friendly, real-time decision-support systems accessible to growers and advisors at operational scales (Filippi et al., 2020; Jones et al., 2021). Model calibration often demands extensive field measurements such as leaf area, soil moisture, and phenology that are impractical for commercial-sized operations (Jiang et al., 2004; Filippi et al., 2020).

Existing satellite-based cotton forecasting pipelines often apply methods developed for grain crops without adaptation to cotton's unique physiology and phenology (Becker-Reshef et al., 2010; Kross et al., 2015). For example, accurate in-season cotton yield forecasts require capturing physiological processes that directly build lint yield such as biomass accumulation, canopy water stress, and the timing of flowering, boll set, and boll opening rather than relying solely on overall "greenness" (Constable & Bange, 2015; Pettigrew, 2008). These yield-driving traits are tightly linked to intercepted photosynthetically active radiation, radiation-use efficiency, and harvest index, all of which vary dynamically throughout the growing season (Constable & Bagne, 2015; Lokhande & Reddy, 2014). Moreover, current satellite-based yield studies often use data from isolated time points or narrow windows within the season. It creates inconsistency and confusion for end users attempting to reproduce these methods (Filippi et al., 2020; Pipia et al., 2022). Since yield is the cumulative outcome of season-long interactions among environment, management, and genetics, forecasting requires integrating temporal profiles rather than snapshots (Fatholouloumi et al., 2026). Therefore, a streamlined workflow that continuously incorporates satellite observations as the crop grows and develops is essential for robust in-season forecasting (Filippi et al., 2020; Zhao et al., 2024).

Recent advances in Artificial Intelligence (AI), particularly deep learning and time-series modeling, provide new opportunities to capture the dynamic interactions among crop growth, environment, and management practices from high-temporal satellite imagery. Unlike traditional statistical approaches that often rely on single-date observations or manually selected features, deep learning models can learn complex nonlinear relationships and temporal dependencies directly from sequential data collected throughout the growing season. Time-series architectures such as Long Short-Term Memory (LSTM) networks, encoder-decoder frameworks, and transformer-based models are especially well suited for analyzing crop growth trajectories because they can continuously incorporate historical observations while adapting to new incoming data (Hochreiter & Schmidhuber, 1997; Olah, 2015). This enables the models to track crop development, detect growth pattern deviations caused by stress or management practices, and improve forecasting accuracy as the season progresses. The increasing availability of high-frequency satellite imagery, such as PlanetScope's near-daily 3 m SuperDove observations, further enhances the potential of AI-driven forecasting systems by providing consistent temporal coverage capable of capturing rapid phenological transitions and within-field variability in cotton (Kabir et al., 2025). In this study, we propose to leverage near-daily PlanetScope Normalized Difference Vegetation Index (NDVI) observations to develop a temporal cotton yield prediction framework capable of learning season-long crop growth patterns and their relationships with final yield.

Specifically, this research aims to (1) develop a cotton yield prediction model using time-series NDVI measurements from PlanetScope imagery, (2) determine how early in the growing season reliable yield forecasts can be achieved, and (3) integrate the trained yield prediction framework into a crop growth forecasting model (CropCast) to support operational in-season forecasting and decision-making. To achieve these objectives, we propose *Pheno-SST*, a phenology-aware temporal model that combines NDVI sequence representation, multi-head self-attention, and a Mamba selective state-space branch to map within-season crop-growth trajectories to end-of-season cotton lint yield. We further compare *Pheno-SST* against benchmark recurrent, Transformer-based, linear, convolutional, and tree-ensemble baseline models, including Bidirectional long short term memory (BiLSTM), Patch Time Series Transformer (PatchTST) Decomposition Linear (DLinear), temporal convolutional network (TCN), and Random Forest (RF), to determine whether the hybrid attention state-space design improves prediction accuracy, bias, and uncertainty calibration under a blind-year test season evaluation strategy. Subsequent sections provide details on dataset preparation, study field, proposed model architecture and results.

Methods

2.1 Study fields

For this study, we use the data collected from cotton fields located at Driscoll, Texas, USA (27°40'21" N, 97°45'4" W) during the 2021-2024 growing seasons. The field is divided into two blocks and cotton sorghum rotation is followed (Figure 1). Driscoll is in the Southern region of Texas where the summers are relatively hot, and winters are milder. The long-term average annual precipitation is 810 mm, mostly after August marking it a dryland cotton production region. Across the four growing seasons, substantial variability was observed in cotton cultivars, planting dates, and crop duration (Table 1). The growing season length ranged from 120 to 150 days from planting to defoliation, reflecting differences in environmental conditions and crop development dynamics among years. This interannual variability provided an opportunity to evaluate the robustness and temporal adaptability of the forecasting framework across diverse production scenarios (Figure 2). The 2022 growing season accumulated heat units more rapidly than other years, while 2021 exhibited the longest growing duration with more gradual GDD accumulation. Rainfall distribution also varied considerably among years, with 2021 receiving substantially greater cumulative precipitation compared to the relatively drier 2022 and 2024 seasons.

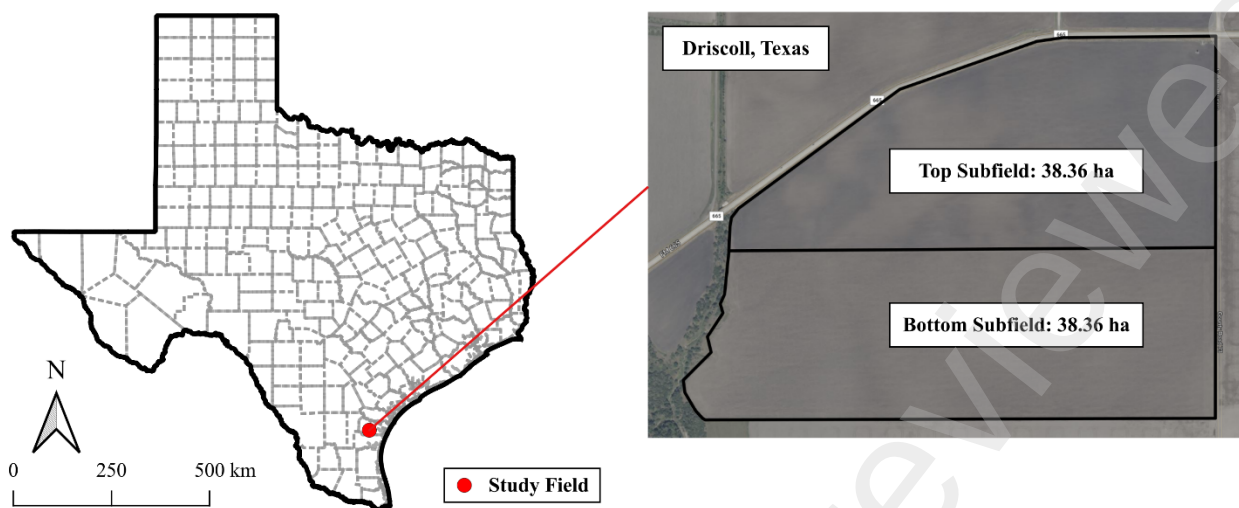


Figure 1. Location of the study area near Driscoll, Texas and an overview of the field. The field is divided into two blocks following a cotton–sorghum rotation system.

Table 1. Crop management details of cotton crop planted during the data collection years

	2021	2022	2023	2024
Cotton cultivar	FM2398GLTP	FM2398GLTP	DP 2239 B3XF	PHY 400 W3FE
Planting date	2021-02-27	2022-03-15	2023-03-05	2024-02-22
Defoliation date	2021-07-27	2022-07-13	2023-07-18	2024-07-11
Harvest date	2021-08-14	2022-08-04	2023-08-01	2024-08-05
Length of the season (planting to defoliation)	150	120	135	132

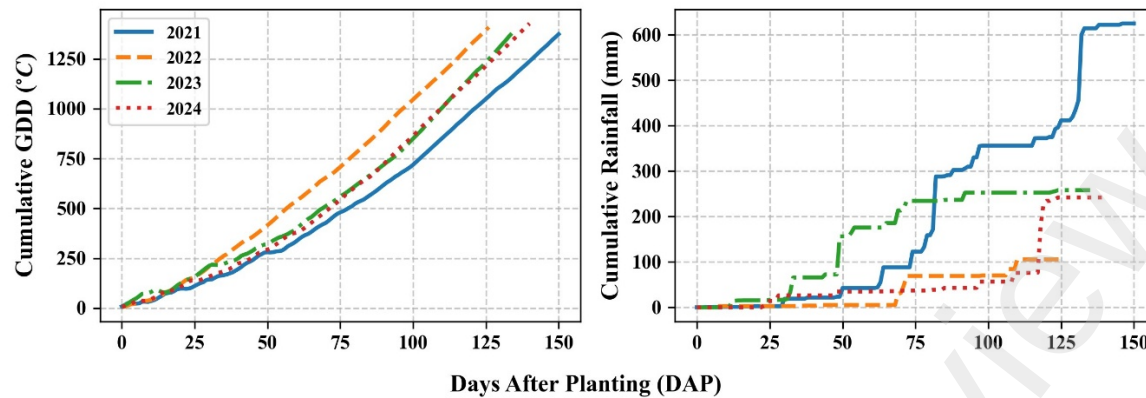


Figure 2. Seasonal weather conditions during the 2021–2024 cotton growing seasons presented as cumulative growing degree days (GDD; left) and cumulative precipitation (right) plotted against days after planting (DAP) for the study area.

2.2. Satellite data acquisition and post processing

We create a vector shapefile delineating the field boundary using QGIS software and use it as the spatial reference for both satellite-image extraction and yield-data gridding. We download harmonized PlanetScope surface reflectance imagery covering each cotton growing season, from planting to defoliation (Table 1). PlanetScope surface reflectance products are orthorectified and radiometrically corrected for temporal monitoring applications, and harmonization further reduces sensor-to-sensor differences across the PlanetScope constellation. We remove images affected by cloud cover, cloud shadow, haze, incomplete field coverage, or other artifacts. We compute normalized difference vegetation index (NDVI) using the red and near-infrared bands. Although several vegetation indices have been proposed to characterize crop growth, canopy structure, and physiological responses (Sarkar et al., 2022; Vidican et al., 2023; Vélez et al., 2023), NDVI was selected as the primary vegetation metric in this study due to its strong association with canopy greenness, photosynthetic activity, biomass accumulation, and final yield (Zhou et al., 2022; Barboza et al., 2023; Caruso et al., 2023; Farias et al., 2023). In addition, the primary objective of this study was to develop and evaluate a temporal forecasting framework rather than perform a comparative assessment of vegetation indices. Therefore, NDVI was selected because of its simplicity, interpretability, and extensive use in agricultural remote sensing applications for monitoring crop growth and seasonal vegetation dynamics across diverse environmental and management conditions. For cotton, NDVI time-series behavior is particularly useful because it reflects canopy establishment, peak vegetative growth, reproductive development, and late-season senescence. However, cotton yield depends not only on maximum greenness but also on the timing and trajectory of growth; therefore, the temporal structure of NDVI was preserved rather than reducing each grid cell to a single seasonal summary value. For the image time series obtained after removing the ones affected by artifacts and cloud, we align the remaining image observations by days after planting to account for differences in planting date and crop-development timing

among years. Subsequently, we generate a 9m x 9m grid within the field and spatially average the obtained NDVI for each grid cell, generating a grid-level NDVI time series for each year (Figure 3). We further smooth the irregular NDVI observations to reduce image noise and residual atmospheric effects while preserving the seasonal crop-growth trajectory. Furthermore, since cotton yield depends not only on the magnitude of canopy greenness but also on the timing of growth, peak canopy development, and senescence, we also compute a first-order derivative of the smoothed NDVI curve to represent the rate of canopy change and include this as an input feature for model learning. To evaluate the effect of including the first order derivative, we also train models without using this as input and evaluate the outcomes.

For training target data, we first inspect and remove missing, duplicate, spatially erroneous, and unrealistic values from the harvest yield observations. We then aggregate the cleaned yield observations to the same 9m grid, and the mean yield within each grid cell is used as the prediction target. The final model input for each grid cell consisted of the smoothed NDVI trajectory and its first-order derivative, while the final harvested yield was used as the response variable. To assess in-season forecasting capability, model inputs were truncated at multiple DAP cutoffs. To assess in-season forecasting capability, model inputs were truncated at multiple days after planting (DAP), and the remaining portion of the growing season was forecasted using CropCAST, a crop growth forecasting framework driven by context aware deep learning model (Math et al., 2026). This approach enabled operational implementation of the trained yield prediction model for in-season forecasting capabilities.

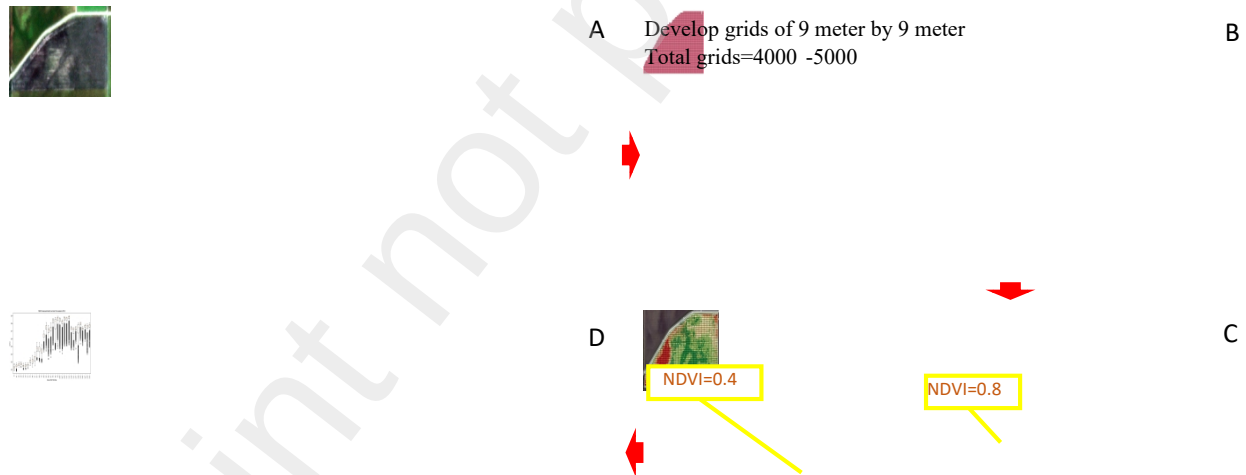


Figure 3. Workflow of extracting Normalized Difference Vegetation Index (NDVI) from the PlanetScope Satellite images (A) Satellite image of the study area. (B) The field was divided into regular grids of 9 m × 9 m, resulting in approximately 4,000–5,000 grid cells for analysis. (C) Spatial distribution of NDVI values across the field, illustrating within-field variability in vegetation vigor (e.g., low NDVI ≈ 0.4 and high NDVI ≈ 0.8). (D) Temporal NDVI dynamics

across the growing season represented as distributions of NDVI values across all grid cells plotted against days after planting (DAP).

2.3. Data preparation – Log normal smoothing and first order derivative

Although Planet Labs' PlanetScope constellation provides near-daily imagery, practical use of satellite data is often constrained by irregular temporal coverage due to cloud contamination, atmospheric conditions, and acquisition limitations. These inconsistencies result in unevenly spaced NDVI observations across the growing season, hindering the accurate assessment of growth trajectories. To fully leverage the temporal information, the irregular NDVI time series was transformed into a regular daily time series through interpolation. In this study, a log-normal function was employed to model and interpolate the NDVI trajectory (Bak et al., 2025). The Log-normal function was selected because it provides a biologically meaningful representation of crop growth dynamics while minimizing the risk of overfitting irregular observations (equation 1). Cotton growth typically follows a characteristic phenological pattern: slow vegetative development during early establishment, followed by a rapid (near linear) increase in canopy expansion and NDVI values. After reaching peak vegetative growth, the crop enters a relatively stable plateau phase. Upon reaching cutout, physiological resources shift toward reproductive development, resulting in reduced vegetative growth rates. As boll development and opening progress, the crop begins senescence, characterized by chlorophyll degradation and a corresponding decline in NDVI values. When approximately 50–60% of bolls are open, defoliant is typically applied, accelerating leaf drop and further reducing NDVI. The log-normal function effectively captures this asymmetric growth–decline pattern, gradual rise, peak, and progressive senescence without overfitting noise from irregularly spaced observations. By fitting this function into seasonal NDVI data, we generated smooth, continuous daily time series.

$$y = y^0 + \frac{a}{x} \cdot \exp \left(-0.5 \cdot \left(\frac{\ln \left(\frac{x}{x_0} \right)}{b} \right)^2 \right) \dots \dots \dots \text{equation 1}$$

such as:

- y = fitted NDVI
- x = days after planting, DAP
- y_0 = baseline NDVI
- a = amplitude / scale
- x_0 = location parameter controlling peak timing
- b = spread/width of the curve

because of the $\frac{a}{x}$ term, this function is asymmetric and usually better for a single-peaked curve that rises and then declines.

2.4. Yield data

Yield data was obtained from JD yield monitor system and is calibrated as described in Lei et al. (2024). Briefly, the total seed cotton yield harvested by the producers and measured by the gin was distributed across the field by calculating the percentage of yield contributed by each point to the total yield. The distribution of cotton yield varied substantially among the four growing seasons (Figure 4). The 2021 season exhibited considerably higher yield values, with most grid cells ranging between approximately 35 and 50 kg per grid, indicating more favorable growing conditions and greater productivity. In contrast, the 2022, 2023, and 2024 seasons showed lower and relatively narrower yield distributions, with peak frequencies concentrated between approximately 15 and 25 kg per grid.

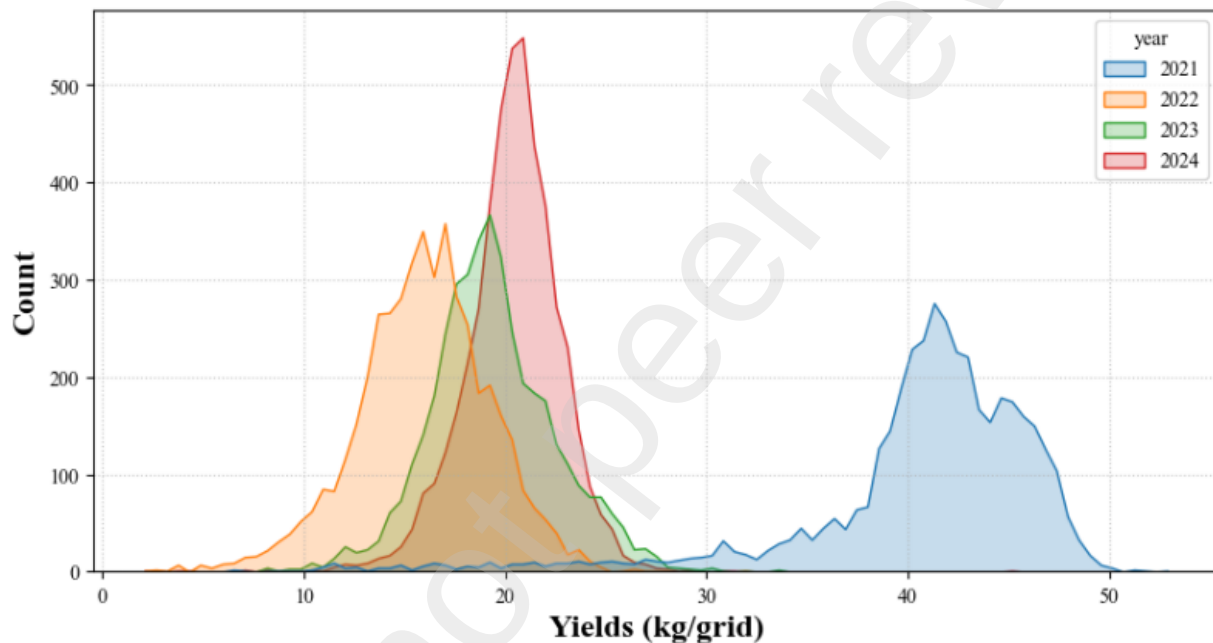


Figure 4. Distribution of cotton yield across grids for the 2021–2024 growing seasons

2.5. Model architecture

We propose a temporal deep learning model, referred to as *Pheno-SST* for in-season cotton yield forecasting using PlanetScope-derived NDVI time series. The model is a phenology-aware sequence-to-sequence Transformer model that is developed to map within-season NDVI trajectories to end-of-season cotton lint yield under cross-regime generalization constraints. The model is deliberately designed around two complementary inductive biases that recur in agronomic time-series, (i) *a short-range, locally coherent component*, for example the inflection separating rapid canopy expansion from peak biomass, and (ii) *a long-range, slowly accumulating component*, for example the integrated thermal and radiative dose between emergence and first square. These two regimes are modelled in parallel by a pre-LayerNorm multi-head self-attention branch and a selective state-space (Mamba) branch (Gu & Dao, 2023). These are recombined

through a learnable gated convex fusion before being decoded by a small multi-layer perceptron yield head. The entire network contains 548,482 trainable parameters, takes a two-channel NDVI sequence of length 127 days-after-planting (DAP) as input, and produces a single scalar lint-yield estimate per field. The model architecture and its implementation workflow is shown in Figure 5. The input to the first model training consists of grid-level PlanetScope NDVI time series aligned by DAP. Later, the first-order derivative of the smoothed NDVI curve was also included as an input feature. Outputs from the Transformer and state-space branches were combined through a gating mechanism and passed to a regression head to estimate final cotton yield. The model therefore acts as a hybrid temporal architecture: the Transformer branch captures global temporal dependencies, while the state-space branch captures continuous crop-growth dynamics.

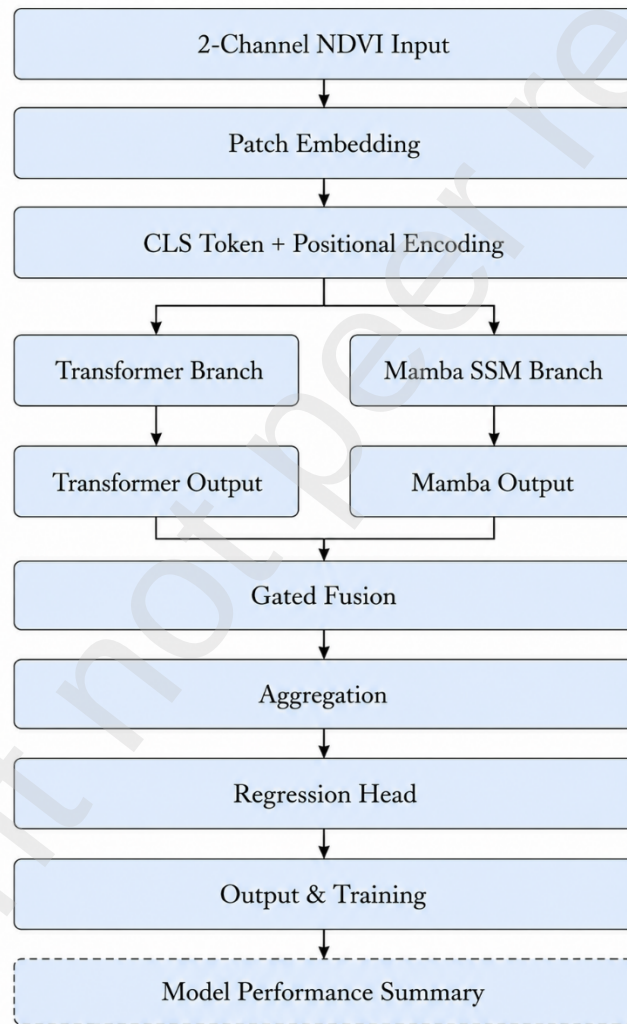


Figure 5. Overview of the *Pheno-SST* network architecture developed for cotton yield prediction using PlanetScope NDVI time-series data. The framework integrates transformer encoder and Mamba state-space model (SSM) branches to capture both short-range and long-range temporal dependencies in crop growth dynamics. Processed NDVI sequences were tokenized and encoded

before feature fusion and regression-based yield prediction, while Monte Carlo dropout was incorporated during inference to estimate prediction uncertainty.

The model input is a tensor $X \in \mathbb{R}^{B \times T \times C}$, consisting of a two-channel temporal sequence with dimensions $B \times T \times C$, where B is the batch size, $T = 127$ represents the number of DAP-aligned time steps, and $C = 2$ represents the input channels. The first channel contains the per-sample z-scored NDVI trajectory, while the second channel contains the first-order derivative of the smoothed NDVI curve. The temporal axis spans 127 DAP, a window that brackets emergence through harvest maturity for the cotton cultivars considered, and the channel axis carries two physically distinct views of the phenological signal. For each field j , we standardise the raw NDVI trajectory using its own seasonal mean and standard deviation for channel 0, as shown in equation (2), where $\tilde{x}_{j,t}^{(0)}$ refers to the channel input, μ_j and σ_j represent the field mean and standard deviation, respectively, and $\text{NDVI}_{j,t}$ refers to the NDVI value for field j at time t . This per-sample normalization removes baseline-greenness differences arising from soil background, cultivar, irrigation regime and atmospheric correction residuals, while preserving the shape of the trajectory, the agronomically informative quantity for yield formation. For channel 1, we retain the derivative in its unstandardized form as shown in equation (3), which sharpens rate-of-change features like green-up slope, peak-plateau duration, and senescence rate, which are masked by per-sample z-scoring of the level channel. The two channels are stacked along the channel axis and presented jointly to the patch embedding, so that the network learns coupled magnitude-derivative phenological signatures rather than treating level and slope independently.

$$\tilde{x}_{j,t}^{(0)} = \frac{\text{NDVI}_{j,t} - \mu_j}{\sigma_j}$$

(2)

$$\Delta \text{NDVI}_{j,t} = \text{NDVI}_{j,t} - \text{NDVI}_{j,t-1}$$

(3)

Before temporal encoding, the input sequence is divided into overlapping patches of length $L_p=10$ days with stride $s=5$ days, producing 24 patches per sample. Each patch contains both NDVI channels and is projected into a 128-dimensional embedding shown in equation (4). Each patch contains local information about short-term crop development, and the overlap between neighboring patches helps preserve continuity in the crop-growth trajectory. The flattened patch vectors are passed through a shared linear embedding layer that maps each patch to a 128-dimensional latent representation. This patch embedding reduces the effective sequence length and allows the model to process local crop-growth segments rather than individual daily values. In this case, the 127-day sequence

$$z_i = W_p \text{vec}(X_i) + b_p, \quad i = 1, \dots, 24.$$

(4)

A learnable classification (CLS) token is added to the patch sequence, and learned positional embeddings are used to preserve the absolute DAP timing. Patch embedding reduces the sequence length, groups approximately 10-day phenological windows into meaningful units and helps smooth high-frequency NDVI noise from residual clouds, bidirectional reflectance effects, or view-angle differences. The token sequence is processed in parallel by the Transformer and Mamba branches. The encoded token sequence is processed by two parallel temporal-learning branches. The first branch is a Transformer encoder consisting of two layers, eight attention heads, pre-LayerNorm, GELU activation, and a feed-forward hidden dimension of 256. This branch is included to capture relationships among different parts of the growing season. Multi-head self-attention allows the model to learn which temporal patches are most informative for yield prediction and how early-, mid-, and late-season canopy conditions interact. The second branch is a Mamba state-space model branch consisting of two layers with a state dimension of 16, local convolution width of 4, and expansion factor of 2. This branch is included to model sequential crop-growth dynamics efficiently. While the Transformer branch captures global relationships among temporal patches, the Mamba branch provides a complementary representation of the ordered NDVI trajectory. The branch outputs are combined by a learned scalar gate shown in equation (5), where h_{TR} is the Transformer representation, h_{MB} is the Mamba representation, and g is the learned gating weight. A temperature parameter is used in the gate calculation to smooth the branch-weighting behavior. The gated fusion allows the model to learn the relative contribution of the attention-based and state-space representations rather than fixing this balance manually. The training target is observed lint yield at the field or grid-cell level. Yield values are standardized using the mean and standard deviation of the training set shown in equation (6).

$$h_{fused} = g h_{TR} + (1 - g) h_{MB}, \quad 0 < g < 1.$$

(5)

$$y_{std} = \frac{y - \mu_y}{\sigma_y}, \quad \hat{y} = \hat{y}_{std} \sigma_y + \mu_y$$

(6)

The model predicts standardized yield during training, and predictions are converted back to physical yield units after inference. After fusion, a field-level seasonal embedding is created by combining the fused classification token with the mean of the fused patch tokens. This representation contains both a learned summary of the most informative temporal features and a stable average description of the season. A three-layer multilayer perceptron maps this 128-dimensional representation to a single standardized yield estimate. The model is trained end-to-end using Huber loss as loss function with threshold $\delta=1.0$. Huber loss behaves like squared error for small residuals but reduces the influence of unusually high- or low-yield observations, which is useful for agricultural yield data affected by localized stress, management differences, or weather extremes. The 2021-2023 growing seasons were used for model training and calibration, and the 2024 season was held out as a blind test set. No truncation augmentation was used; each sample contained the full 127-DAP sequence.

To evaluate whether the proposed *Pheno-SST* architecture improves yield prediction beyond established sequence learning and machine learning approaches, we compared it with BiLSTM, PatchTST, DLinear, RF, and TCN. These baselines represent recurrent sequence learning, patch-based Transformer learning, linear time-series modeling, tree-ensemble regression, and convolutional temporal modeling, respectively. BiLSTM was included as a recurrent baseline based on LSTM and bidirectional LSTM sequence modeling (Hochreiter and Schmidhuber, 1997; Graves and Schmidhuber, 2005), PatchTST as a patch-based Transformer baseline (Nie et al., 2023), DLinear as a simple linear time-series forecasting baseline (Zeng et al., 2023), RF as a non-deep-learning ensemble benchmark (Breiman, 2001), and TCN as a convolutional sequence-modeling baseline (Bai et al., 2018). All models were trained and evaluated using the same PlanetScope-derived NDVI time-series inputs, DAP alignment, yield targets, training years, and blind 2024 test dataset. This comparison was designed to test whether the hybrid Transformer–Mamba structure provides added value for modeling cotton yield from phenological trajectories. The Transformer branch builds on multi-head self-attention (Vaswani et al., 2017), while the Mamba branch uses selective state-space sequence modeling (Gu and Dao, 2023). Therefore, comparison with BiLSTM, PatchTST, DLinear, RF, and TCN allows us to assess whether the hybrid Transformer–Mamba structure improves point-prediction accuracy, bias, and uncertainty calibration relative to models based on a single dominant learning mechanism. The next section shows the results obtained using the proposed model and its comparison with the baseline models.

2.6. Model performance evaluation metrics

To evaluate the performance of the developed model under varying experiments, three metrics including Root Mean Square Error (RMSE), mean absolute error (MAE), and bias were chosen, which are calculated as below. Term n denotes the number of data samples, y_i and $\hat{y}_i$ are the observed and estimated yield for the i^{th} sample, and $\bar{y}$ is the mean value of observed yield values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (8)$$

$$Bias = \frac{\sum_{i=1}^n y_i - \hat{y}_i}{n} \quad (9)$$

3. Results

3.1. Seasonal NDVI dynamics and Log normal smoothing

Figure 6 illustrates the temporal dynamics of NDVI across three growing seasons (2021–2023) as a function of days after planting (DAP), integrating both observed data distributions and a log-normal based smoothing approach. NDVI observations exhibit substantial within-date variability, particularly during the mid-season period of rapid canopy development, indicating pronounced heteroscedasticity. The fitted log-normal daily mean effectively captures the seasonal progression of crop growth, representing early-stage low NDVI values, a rapid increase during vegetative growth, peak canopy development, and a gradual decline associated with senescence. The modeling framework, based on a logarithmic transformation of NDVI, stabilizes variance and reduces the influence of extreme values, resulting in a smooth and biologically consistent representation of crop development. This demonstrates the effectiveness of a log-normal based approach for robustly characterizing NDVI time series and capturing both crop growth and development dynamics and variability across diverse growing conditions. Similar curve-fitting approaches are widely used in cotton and other crop studies (Palla et al., 2025; Baker et al., 2026) to convert irregular time-series observations into continuous and regular temporal datasets, enabling improved characterization of seasonal vegetation growth and senescence dynamics while minimizing overfitting caused by noisy or unevenly distributed observations.

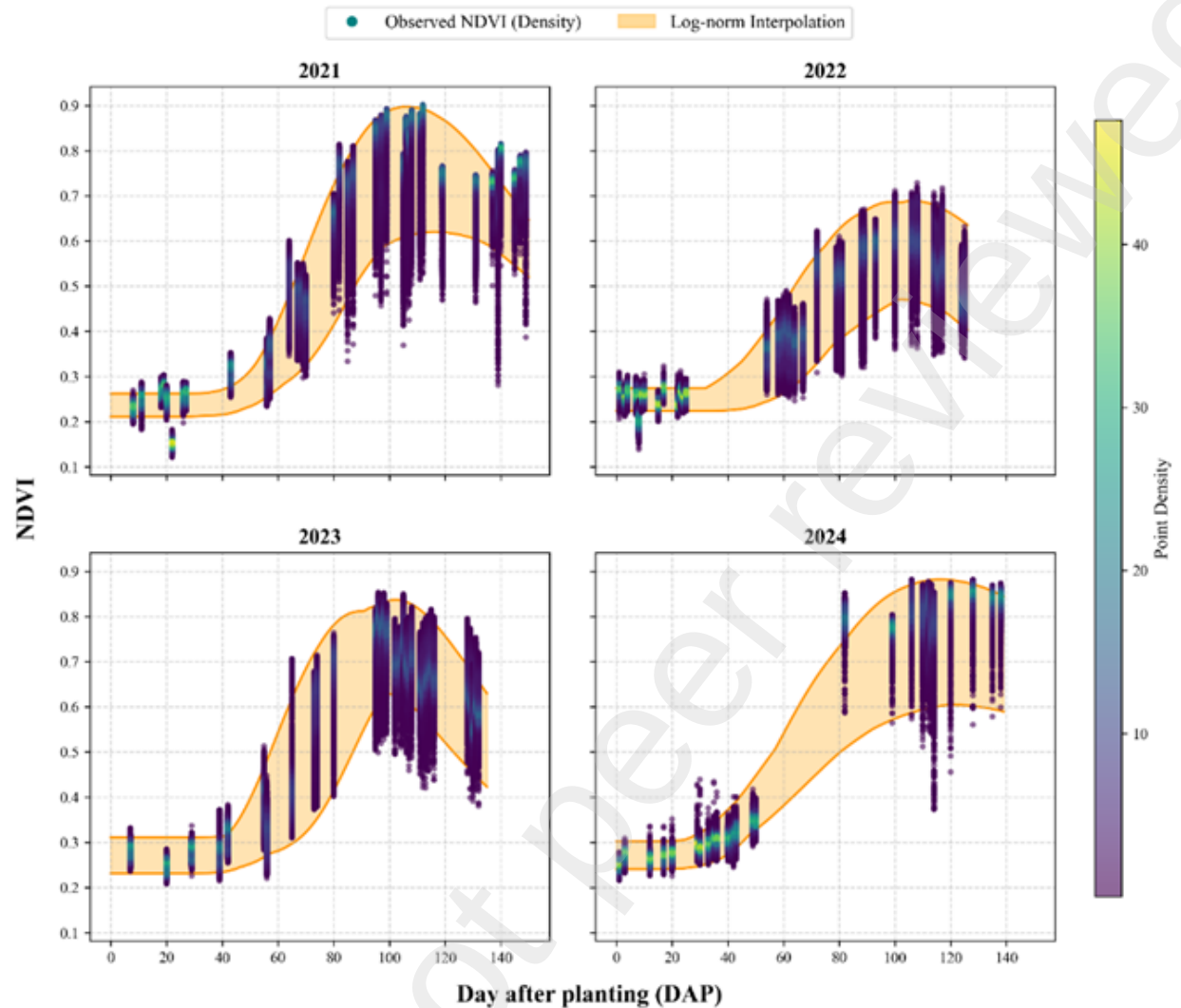


Figure 6. Temporal Normalized Difference Vegetation Index (NDVI) dynamics during the 2021–2024 cotton growing seasons. Colored scatter points represent observed PlanetScope NDVI values with point density, while the orange curves represent fitted log-normal interpolations used to smooth seasonal vegetation trajectories across days after planting (DAP).

3.2. Summary statistics of seasonal NDVI and yield across years

Table 2 summarizes interannual variability in canopy dynamics, vegetation indices, and yield-related metrics from 2021 to 2024. Substantial differences were observed across years, reflecting variability in environmental conditions and crop performance. The mean peak NDVI ranged from 0.597 in 2022 to 0.868 in 2021, indicating comparatively reduced canopy development in 2022. This is further supported by lower mean area under the curve (AUC) values in 2022 (51.29) relative to other years, suggesting reduced cumulative canopy growth. In contrast, 2024 exhibited the highest mean AUC (66.02) and elevated mean NDVI at the 0.95 quantile (40.16), indicating stronger canopy persistence and vigor.

Temporal dynamics also varied notably, with mean peak time occurring earliest in 2023 (96.36 days) and latest in 2024 (120.52 days), suggesting delayed crop development in 2024. Variability in peak timing was highest in 2022 (standard deviation = 5.54), indicating less synchronized crop growth across fields. Yield patterns reflected these differences, with 2021 showing the highest mean yield (41.93) and maximum yield (51.65), while 2022 exhibited a substantial decline in mean yield (16.02), likely associated with reduced canopy development. Correlation between peak NDVI and yield was strongest in 2024 (0.742), suggesting improved predictive relationships between canopy vigor and yield under those conditions. Overall, the results highlight the sensitivity of canopy development metrics and yield to interannual variability, emphasizing the importance of integrating temporal vegetation dynamics into in-season crop monitoring and forecasting frameworks.

Table 2: Summary statistics of log-normal Normalized Difference Vegetation Index (NDVI) curve parameters and yield distribution across the 2021–2024 growing seasons, including peak NDVI characteristics, area under the curve (AUC), yield variability, and correlations (r) between peak NDVI and yield. Std: Standard Deviation

Year	Sample Size	Mean Peak	Mean Plant	Mean Peak Time	Std Peak Time	Mean AUC	Std AUC	Mean Yield	Std Yield	Max Yield	Min Yield	r Peak Yield	Mean NDVI (0.95)	Std NDVI (0.95)
2021	3727	0.868	0.238	107.422	1.788	62.561	1.921	41.934	3.540	51.649	31.231	0.398	35.541	1.546
2022	4492	0.598	0.253	97.236	5.538	51.294	2.053	16.015	3.040	24.226	7.706	0.674	33.321	1.692
2023	3976	0.757	0.260	96.365	3.714	59.790	2.417	19.314	2.855	26.876	11.716	0.627	37.539	2.301
2024	4593	0.827	0.265	120.517	4.258	66.018	1.725	20.445	2.041	25.837	15.069	0.742	40.161	0.915

3.3 Performance of the proposed model for yield prediction using temporal NDVI series

The performance of the proposed *Pheno-SST* model was first evaluated using NDVI magnitude alone as the only input feature (Figure 7). All deep learning models showed high performance during training, with validation R^2 values close to 0.85. However, their performance dropped sharply when applied to the test year. This indicates poor generalization across years. For *Pheno-SST*, test performance was very low, with RMSE = 7.15, and MAE = 6.80. The model showed a strong negative bias (−6.80), meaning it consistently underestimated yield. PatchTST and BiLSTM showed similar patterns. RF showed a large positive bias (+12.57), while TCN and DLinear performed the worst. Figure X shows that predictions collapsed into a narrow low-yield range. This resulted in weak or negative agreement with observed yield. After adding the first-order derivative of NDVI, model performance improved significantly (Figure 8). *Pheno-SST* achieved a score of 2.89 and bias close to zero (+0.30). Other models did not show similar improvement.

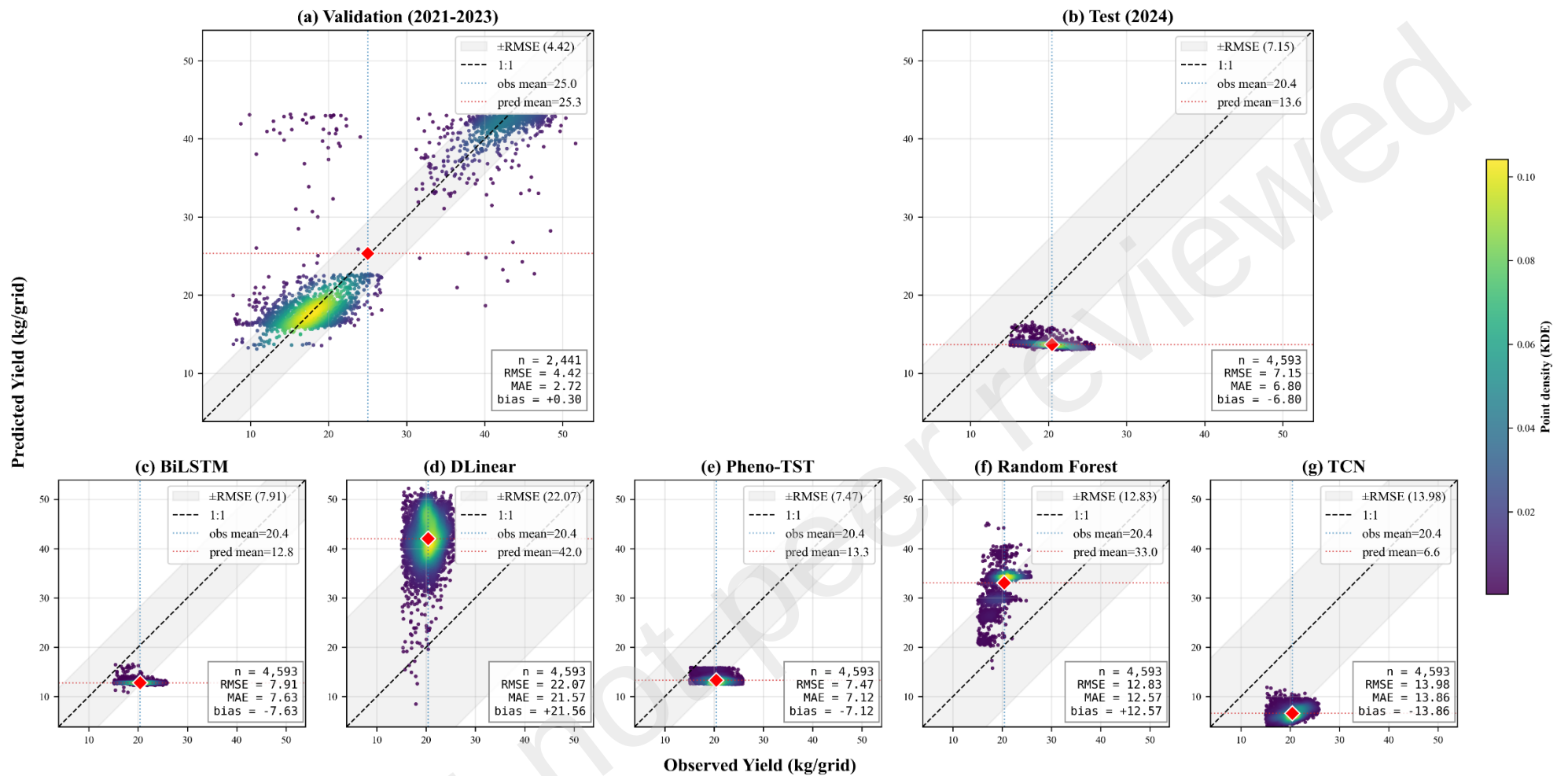


Figure 7: Predicted versus observed cotton yield for the proposed model and baseline models. Panels (a) and (b) show validation (2021–2023) and blind test (2024) results, respectively, for the proposed model, while panels (c)-(g) show blind-test results for the benchmark models: BiLSTM, DLinear, Pheno-TST, Random Forest, and TCN. The dashed black line represents the 1:1 relationship, and the gray shaded band indicates the $\pm$ RMSE envelope. Blue vertical and red horizontal dotted lines denote the mean observed and predicted yields, respectively, with the red diamond marking their intersection. Point color represents kernel density estimate (KDE)-based sample density. Insets report the sample size (n), RMSE, MAE, and bias for each panel. The proposed model shows the closest agreement with the 1:1 line and the most favorable overall performance on the 2024 blind test.

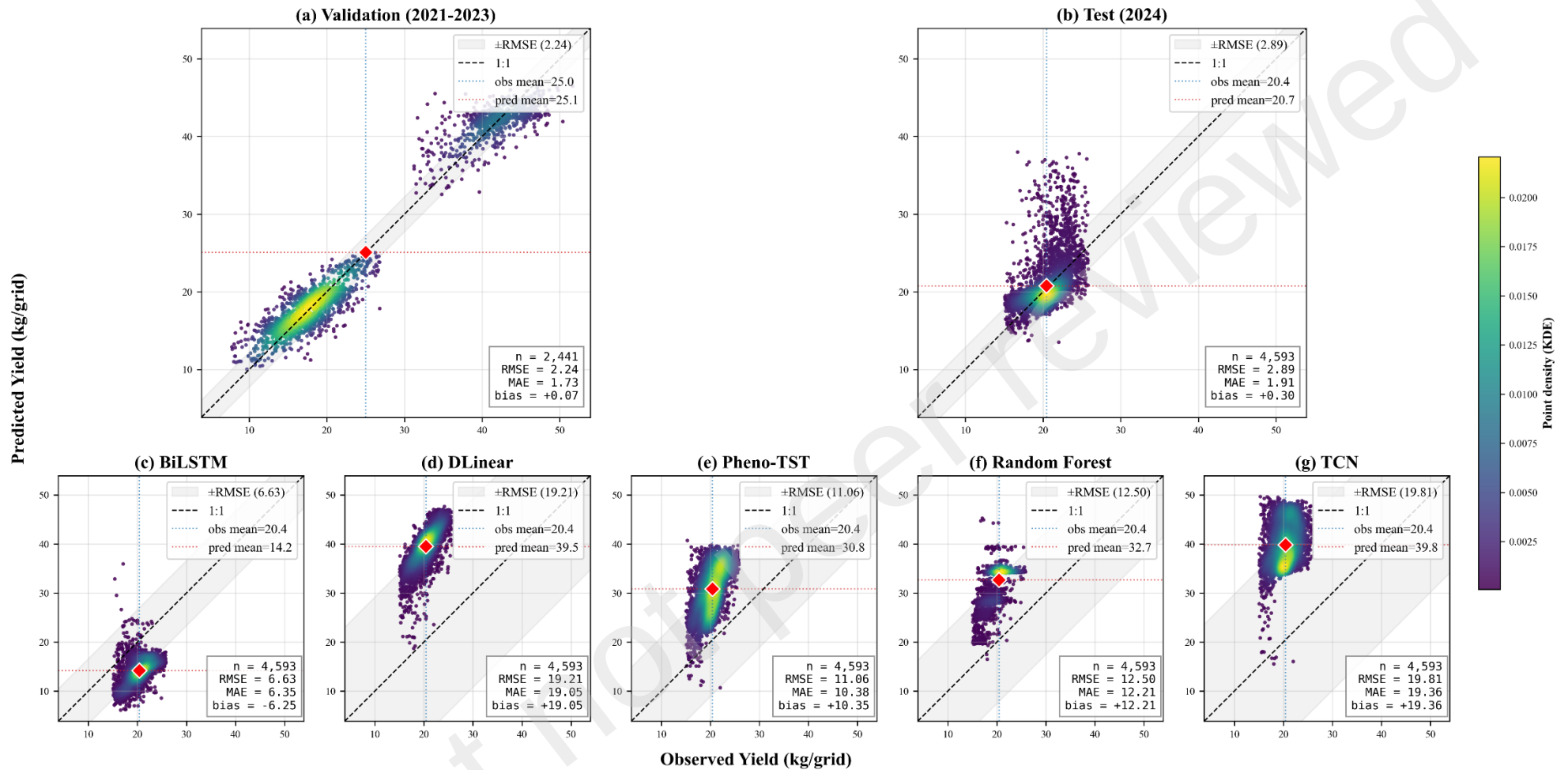


Figure 8: Comparison of yield prediction performance between the proposed model and competing methods using validation and blind-test datasets. Panels (a) and (b) present validation (2021–2023) and blind test (2024) performance for the proposed model, while panels (c)–(g) summarize blind-test predictions for BiLSTM, DLinear, Pheno-TST, Random Forest, and TCN, respectively. Scatter plots compare predicted and observed cotton yield at the grid level. The dashed 1:1 line indicates perfect agreement, and the gray shaded region corresponds to $\pm$ RMSE. Vertical blue and horizontal red dotted lines indicate mean observed and predicted yields, respectively. Color intensity reflects KDE-based point density, and inset boxes provide n , RMSE, MAE, and bias. Relative to the benchmark models, the proposed model yields lower prediction error and reduced systematic bias on the 2024 blind-test set.

3.4 Uncertainty analysis

Uncertainty in yield prediction was evaluated using conformal prediction intervals (CPIs) (Figure 9). When models were trained using NDVI magnitude alone, all models showed poor calibration. The observed coverage was much lower than the expected confidence levels. This indicates overconfident predictions. After including the NDVI derivative, uncertainty estimates improved for *Pheno-SST*. The model achieved 89.6% coverage for the 90% interval and 92.5% coverage for the 95% interval. These values are close to the expected levels. The prediction intervals were symmetric and centered around the predictions. Interval width increased at yield extremes and remained smaller in mid-range conditions. Other models still showed poor calibration even after adding derivative features.

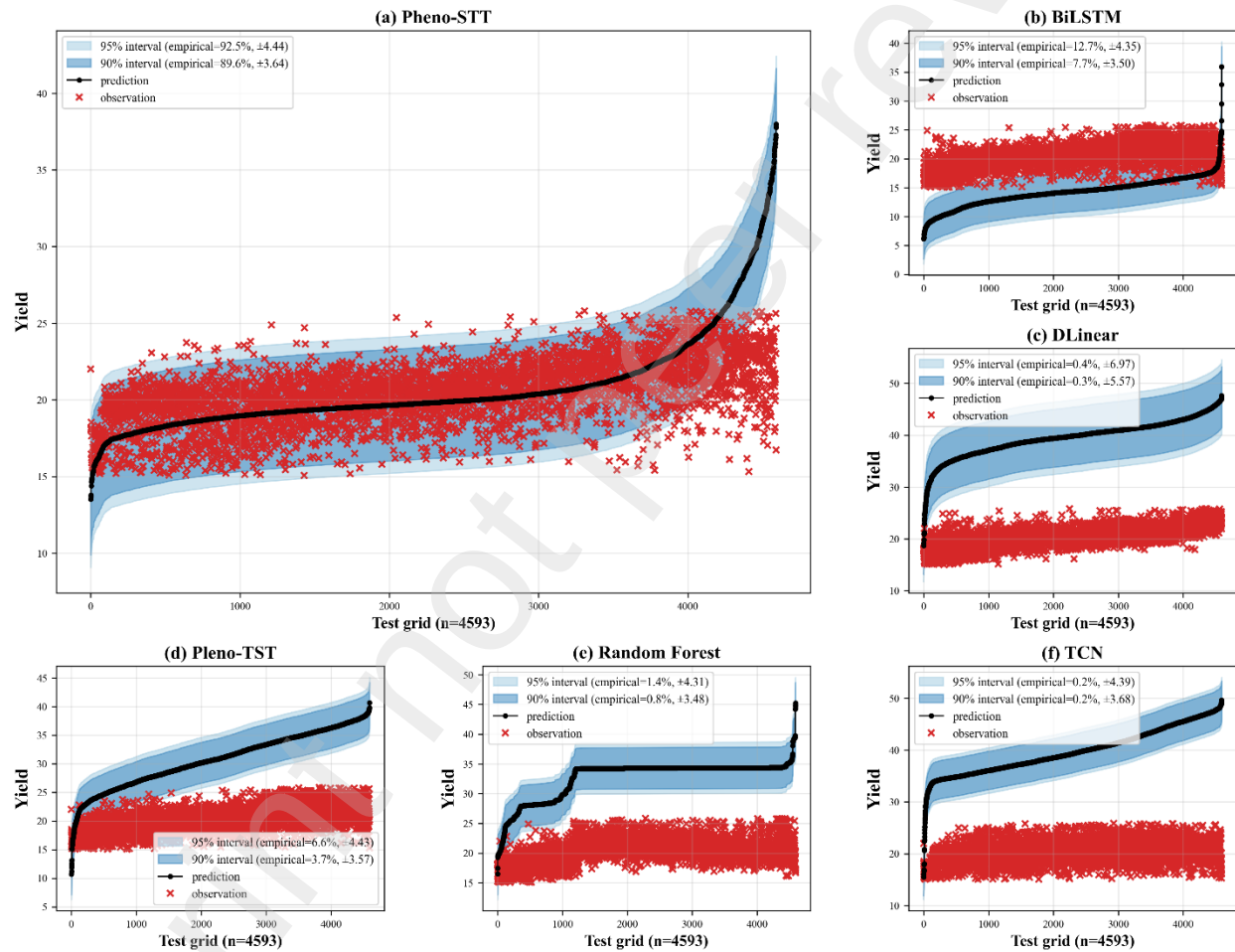


Figure 9: Prediction intervals and uncertainty calibration for the proposed model and baseline models on the 2024 blind-test dataset. Panels show sorted grid-level predictions for (a) Pheno-STT, (b) BiLSTM, (c) DLinear, (d) Pheno-TST, (e) Random Forest, and (f) TCN. Black curves denote ordered predictions, red crosses indicate observed yields, and the dark and light blue shaded regions represent the 90% and 95% prediction intervals, respectively. Legends report empirical interval coverage and average interval width. The proposed model provides well-calibrated uncertainty bounds with empirical coverage close to the nominal levels, whereas several

benchmark models exhibit substantial over- or under-coverage and wider or poorly centered intervals.

3.5 Integrate trained yield prediction model with CropCAST for in-season yield forecasting

In-season yield prediction is critical for supporting timely crop management, irrigation scheduling, harvest planning, and marketing decisions before the end of the growing season. Therefore, the trained *Pheno-SST* yield prediction model was integrated with CropCast to streamline operational in-season forecasting by enabling continuous crop growth prediction and early-season yield estimation using forecasted vegetation dynamics. The integrated framework was evaluated for in-season yield prediction at multiple growth stages (57, 81, 105, and 117 DAP) by truncating the observed input time series at each stage and forecasting the remaining portion of the season using CropCast. Prediction accuracy varied substantially across growth stages (Table 3). At 57 DAP, the model produced large errors and strong overestimation tendencies (bias = +12.06; RMSE = 14.96). Performance further deteriorated at 81 DAP, resulting in the highest prediction error (bias = +22.26; RMSE = 22.50), indicating limited predictive stability during the early-to-mid vegetative growth period. Prediction accuracy improved considerably at 105 DAP, where RMSE decreased to 6.54 and bias was reduced to +3.74. By 117 DAP, predictions became substantially more stable, with near-zero bias (−0.97) and low prediction error (RMSE = 4.08), approaching the performance achieved using full-season observations. Model performance during the early growth stages remained weak and spatially dispersed, suggesting limited availability of stable phenological information for reliable early season yield forecasting.

Table 3: In season yield prediction performance metrics using CropCast. DAP: Days After Planting, MAE: Mean Absolute Error, RMSE: Root Mean Square Error

DAP	Predicted mean	Bias	MAE	RMSE
57	32.50	+12.06	12.26	14.96
81	42.71	+22.26	22.26	22.50
105	24.18	+3.74	6.54	6.54
117	19.47	−0.97	4.08	4.08

4. Discussion:

A central gap identified in this work is that existing satellite-based yield forecasting systems often treat yield prediction as a snapshot problem, using single-date or narrow-window observations rather than integrating the full temporal trajectory of crop development (Filippi et al., 2020; Fatholouloumi et al., 2026). This study addresses that gap by proposing *Pheno-SST*, a hybrid deep learning architecture that learns from the entire season-long NDVI trajectory and its rate of change. The results validate this approach and, more importantly, reveal why temporal trajectory

modeling is not merely a technical refinement but a fundamental requirement for robust cotton yield forecasting.

It is observed that NDVI magnitude alone, even when drawn from near-daily PlanetScope imagery, fails to generalize across growing seasons. When models were trained using only NDVI as input, all architectures, including BiLSTM, PatchTST, and RF, showed high performance during training but collapsed on the 2024 test year (Figure 7). *Pheno-SST* under this condition produced an RMSE of 7.15 and a strong negative bias of -6.80 . The failure may however reflect a deeper agronomic reality too apart from the model performance or data scarcity and quality. Similar NDVI magnitudes can arise from very different physiological trajectories. A slowly rising NDVI could indicate delayed growth or early-season stress. A plateau could reflect peak vigor. Learning from greenness alone without knowing the rate of change of crop growth governing flowering, boll set, and stress-driven canopy dynamics, it is difficult to model the crop growth curve and relate yield (Constable & Bange, 2015; Pettigrew, 2008). Adding the first-order derivative resolved this ambiguity substantially. After including the derivative, *Pheno-SST* achieved an RMSE of 2.89 and a bias of only $+0.30$ (Figure 8). However, *Pheno-SST* still outperformed the baseline models, indicating that the benefit of derivative features as input is utilized efficiently by the proposed architecture. This finding aligns with recent work showing that hybrid or transformer-based architectures outperform conventional recurrent models for agricultural time series because they can learn relationships across growth stages rather than treating each time step in isolation (Du et al., 2025; Lal et al., 2026).

Model performance on the 2024 test data deserves careful interpretation. Summary statistics in Table 2 reveal that 2024 was distinctly different from the three training years. Mean peak NDVI in 2024 was 0.827, while in 2022 it dropped to 0.598. Mean peak timing in 2024 occurred at 120.52 DAP, substantially later than in 2021 (107.42), 2022 (97.24), and 2023 (96.37). The AUC in 2024 was the highest across all years (66.02), and the correlation between peak NDVI and yield was also highest in 2024 (0.742). These metrics collectively confirm that 2024 represented a growing season with delayed, extended canopy development relative to training years. Despite this, *Pheno-SST* produced well-calibrated predictions on the 2024 full-season data (Figure 8). RMSE scores are low and the model generalized to a new season that was statistically distinct in its phenological timing and canopy behavior. This demonstrates that the model learned transferable representations of crop growth dynamics rather than memorizing year-specific patterns. The Transformer branch's ability to attend to different periods in the seasonal trajectory, and the Mamba branch's capacity to model gradual temporal transitions, together allowed the model to recognize yield-driving patterns even when the timing of those patterns shifted. This highlights distribution shift robustness, a property rarely demonstrated in agricultural deep learning studies and arguably more valuable than peak accuracy on a held-out fold from the same year (Meroni et al., 2021; Betew et al., 2025).

The uncertainty analysis through conformal prediction intervals reinforces the observation that under NDVI-only input, all models showed poor calibration, with observed coverage far below nominal confidence levels, indicating overestimated but incorrect predictions. After adding the derivative, *Pheno-SST* achieved 89.6% coverage for the 90% interval and 92.5% for the 95% interval. These values closely match expected levels, meaning the uncertainty estimates were honest. The adaptive width of the intervals is agronomically meaningful: wider intervals appear at yield extremes, reflecting the model's recognition that yield under stress or unusual growth conditions is inherently harder to predict. Narrow intervals in mid-range yields align with agronomic knowledge that moderate, stress-free growing seasons produce more predictable outcomes (Constable & Bange, 2015). The ability to produce calibrated, distribution-free uncertainty intervals is a feature that most agricultural deep learning models fail to achieve, and it has direct implications for practical decision-making. Growers, traders, and policymakers rely on yield forecasts not just for the point estimate but for the confidence surrounding it (Meroni et al., 2021; Betew et al., 2025). An overconfident wrong forecast can be more damaging than an uncertain correct one. This limitation has been noted across the literature on operational forecasting systems (Filippi et al., 2020; Lal et al., 2026), and this study moves directly to address it through distribution-free conformal coverage.

The in-season forecasting results reveal a pattern that is physiologically interpretable. At 57 DAP, bias was +12.06 and RMSE was 14.96. At 81 DAP, errors worsened further, with bias rising to +22.26 and RMSE to 22.50. Accuracy improved substantially at 105 DAP, and by 117 DAP, bias was near zero (−0.97) and RMSE fell to 4.08. The poor performance at 57 and 81 DAP is consistent with what is known about cotton physiology. During early vegetative stages, NDVI primarily reflects canopy expansion and chlorophyll content, and yield potential at this point is still largely undetermined. The large overestimation at 81 DAP may be attributed to rapid canopy growth produces high NDVI derivatives, and the model interprets vigorous vegetative growth as a signal of high yield. Additionally, due to the limited data available during the linear growth stage, CropCast overestimated the NDVI forecast and with the addition of data collected at 105 DAP, the CropCast stabilized and accurately forecasted the rest of the season (Math et al., 2026). However, in cotton, reproductive success depends on boll set and retention, which are governed by temperature, water availability, and carbohydrate partitioning rather than vegetative vigor alone (Pettigrew, 2008; Constable & Bange, 2015). The decoupling between vegetative greenness and reproductive output is a fundamental characteristic of cotton that distinguishes it from deterministic cereal crops (Constable & Bange, 2015). The improvement after 105 DAP coincides with the onset of boll development, when canopy signals begin to reflect reproductive investment and stress more directly. By 117 DAP, most yield-determining events have occurred, and NDVI trajectories encode cumulative stress, resource allocation, and senescence progress (Pettigrew, 2008). The attention mechanism within the Transformer branch supports this interpretation: attention weight distributions shifted toward mid- and late-season time steps as the season progressed, confirming that the model was learning phenologically meaningful associations rather than fitting arbitrary correlations. Similar behavior has been reported in transformer-based

agricultural models, where attention identifies critical growth periods (Du et al., 2025; Lal et al., 2026).

The 2024 season did not follow the canonical NDVI trajectory observed in 2021 through 2023. In other (usual) years, NDVI rises during vegetative growth, plateaus near peak canopy development, then declines as senescence and defoliation proceed. In 2024, heavy rainfall occurred after the NDVI had begun its senescence decline. This rainfall triggered renewed vegetative growth, causing NDVI to increase again before eventually declining. This anomalous trajectory reflects a fundamental characteristic of cotton: it is not a true annual crop, but a perennial species managed as an annual (Constable & Bange, 2015; Jabran et al., 2019). Under favorable late-season conditions, cotton can resume vegetative growth at the expense of boll retention, a phenomenon known in agronomy as rank growth or late-season vegetative rebound (Constable & Bange, 2015). The CropCast forecasting framework that projected the remaining NDVI trajectory beyond each DAP cutoff was not trained on this behavior (Math et al., 2026). It predicted a standard senescence trajectory when the actual trajectory was a second peak followed by a late decline. This mismatch propagated into yield predictions at earlier cutoff dates. Importantly, when the full observed NDVI trajectory was used without truncation, the model performed well on 2024 data. This implies that early in-season prediction for 2024 originated from the NDVI forecasting step, not from *Pheno-SST* itself, and improving the CropCast component, specifically its ability to model anomalous late-season NDVI trajectories driven by unusual weather, would directly improve in-season prediction accuracy in irregular years. This is a tractable engineering problem and represents a clear direction for future work.

By building a temporal deep learning framework directly on this imagery, the study shows that satellite data, when combined with the right model architecture, can provide the temporal richness needed to resolve physiologically meaningful canopy dynamics. This is consistent with findings in corn, where near-daily 3 m satellite imagery reduced yield prediction error by 15% relative to coarser satellite inputs by better resolving critical phenological transitions (Li et al., 2022). In precision agriculture, models often fail when deployed across locations, seasons, or cultivars because they were trained on features that are season-specific rather than process-general (Radočaj et al., 2026; Sarkar et al., 2025). The inclusion of the NDVI derivative as a biologically grounded feature, capturing the rate of canopy development, is a physically interpretable solution that does not require additional sensors or crop-specific parameterization. The input feature set used in this study can be replaced and augmented across crops and locations, which may further enhance learning the underlying biological process. The framework, once established, can in principle be adapted to other row crops by substituting NDVI with a relevant vegetation index for the target crop, without changing the model architecture (Fatholouloumi et al., 2026; Li et al., 2026).

The integration of calibrated uncertainty through conformal prediction also addresses a gap in the operational deployment of agricultural AI. Most deep learning yield prediction systems provide only point estimates, which are insufficient for risk-based decisions (Lal et al., 2026; Meroni et al., 2021). This design principle, borrowed from reliable machine learning theory, is rarely applied in crop science but is highly valuable for decision support tools used by growers

and insurers who need to quantify downside risk (Filippi et al., 2020; Lal et al., 2026). As the availability of high-frequency satellite data continues to grow, and as machine learning pipelines in agriculture mature, the ability to pair accurate predictions with honest uncertainty estimates will become a baseline expectation rather than a distinguishing feature (Betew et al., 2025). This study contributes an early and rigorous example of what that looks like in practice.

The study connects to a larger conversation about how big data and AI can be made reliable in agriculture. The field has seen rapid growth in model complexity, with deep learning architectures applied to increasingly large satellite datasets (Betew et al., 2025; Radočaj et al., 2026). However, generalization across environments remains a persistent challenge (Meroni et al., 2021; Filippi et al., 2020). The results here suggest that the path to generalization is not primarily through more data or more complex architectures, but through feature design that encodes biological process rather than statistical appearance, and through model structures capable of integrating information across time at multiple scales simultaneously (Fatholouloumi et al., 2026; Du et al., 2025). *Pheno-SST*'s hybrid design, its physiologically informed input features, and its calibrated uncertainty framework together offer a template for how future agricultural AI systems can be built to learn from heterogeneity in the natural environment rather than being limited by it.

Conclusion

This study demonstrated that accurate, scalable, and interpretable in-season cotton yield forecasting is achievable using near-daily PlanetScope satellite imagery when used with a hybrid deep learning architecture. It is evident that NDVI magnitude alone is insufficient for generalizable yield prediction and including the first-order derivative of the smoothed NDVI trajectory, which encodes the rate and direction of canopy change, helps capture the physiological processes that drive yield. The proposed *Pheno-SST* model, which integrates a Transformer branch for long-range temporal dependencies and a Mamba state-space branch for sequential growth dynamics, outperformed benchmark architectures including BiLSTM, PatchTST, RF, TCN, and DLinear, achieving an RMSE of 2.89 and a near-zero bias of +0.30 on the unseen 2024 test year. Data from 2024 was statistically distinct from training years in its peak timing, canopy persistence, and overall NDVI trajectory, demonstrating robustness to the heterogeneity of natural growing environments. When integrated with CropCast model, in-season forecasting results showed that reliable predictions require the crop to reach approximately ~100 days after planting, aligning with the onset of reproductive-stage dynamics in cotton. The integration of conformal prediction intervals provided well-calibrated, distribution-free uncertainty estimates, a capability rarely demonstrated in agricultural deep learning systems, and one with direct value for grower, trader, and policy-level decision-making. Together, these contributions offer a transferable, operationally viable model for satellite-based yield forecasting that can be adapted across crops, regions, and vegetation indices, representing a meaningful step toward AI-driven precision agriculture on a scale.

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